

Arc length



- 1 a 11.5 units b 19.6 units
- 2 a 10.47 cm b 33.25 mm c 9.42 cm d 6.00 cm
- 3 a $\frac{5\pi}{6}$ radians b $\frac{25\pi}{2}$ cm
- 4 $\left(20 + \frac{25\pi}{3}\right)$ cm
- 5 a 36.8 cm b 16.8 cm
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Sector area

- 6 a 463.10 m² b 313.01 m²
- 7 39 π units²
- 8 a 50 cm² b 729.87 m² c 612.42 m²
- 9 a $\frac{2\pi}{3}$ b 108 π cm²
- 10 a 6 π cm² b 6 π cm² c 6 π cm²



Dimensions from length and area

11

a $r = 11.46$

b $r = 5.09$

c $r = 4.79$

d $r = 14.90$

12

a $\theta = 45.02^\circ$

b $\theta = 334.23^\circ$

c $\theta = 190.99^\circ$

13 $\theta = 1.95$ radians

14

a $\theta = 0.9$ radians

b $\theta = 1.1$ radians

15

a $\theta = 2$ radians

b $\frac{9}{4}\pi^2 \text{ cm}^2$

16

a $\angle JOK = 46^\circ$

b 90 cm^2

17

a $\theta = 1.67$ radians

b $\theta = 96^\circ$

18

a $\theta = 104^\circ$

b 586 m^2

19 $\theta = 1.7$ radians

20

a $\theta = 0.8$ radians

b $l = 16 \text{ m}$

21 $l = 11 \text{ cm}$

Applications

--- 2



23 317.9 m^2

24

a $48\pi \text{ cm}^2$

b $336\pi \text{ cm}^2$

25 $70\pi \text{ cm}$

26

a $10\pi \text{ cm}^2$

b $(5\pi + 8) \text{ cm}$